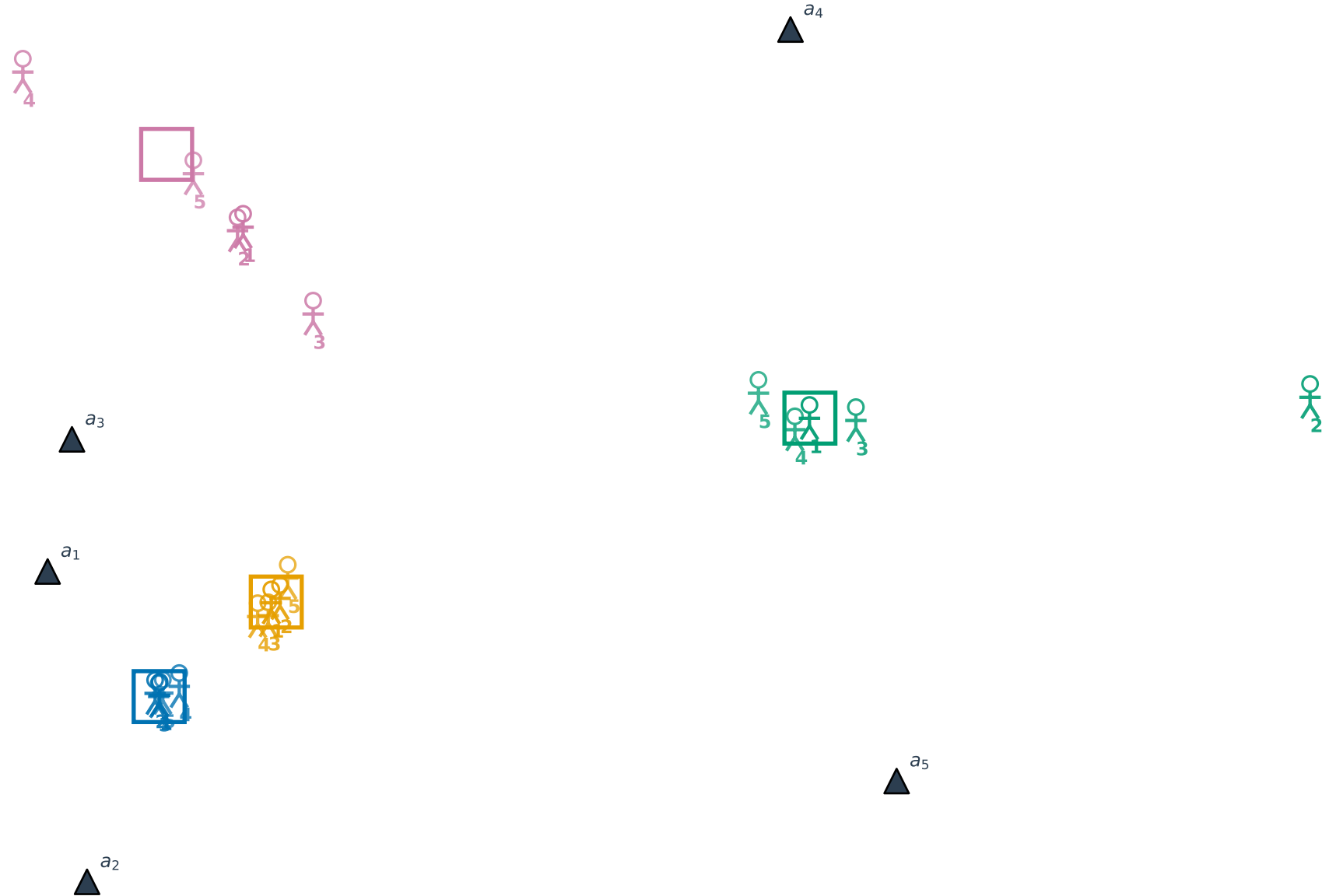
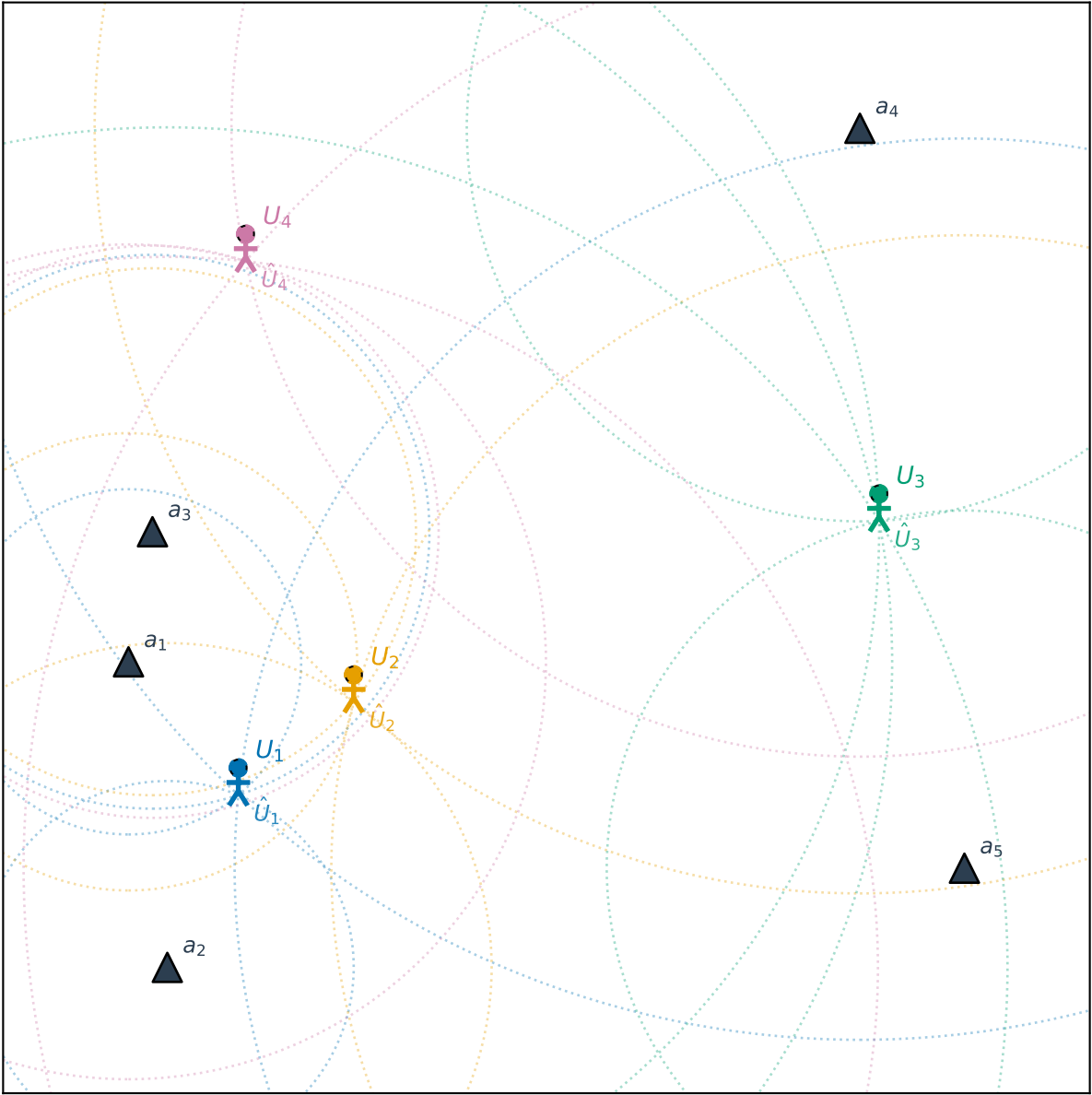


User Localization Drift under Gaussian Noise ($R=4$, $I=5$)



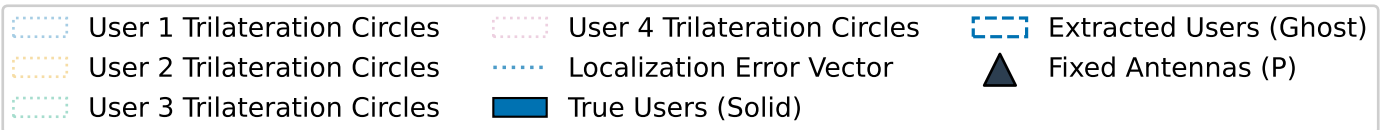
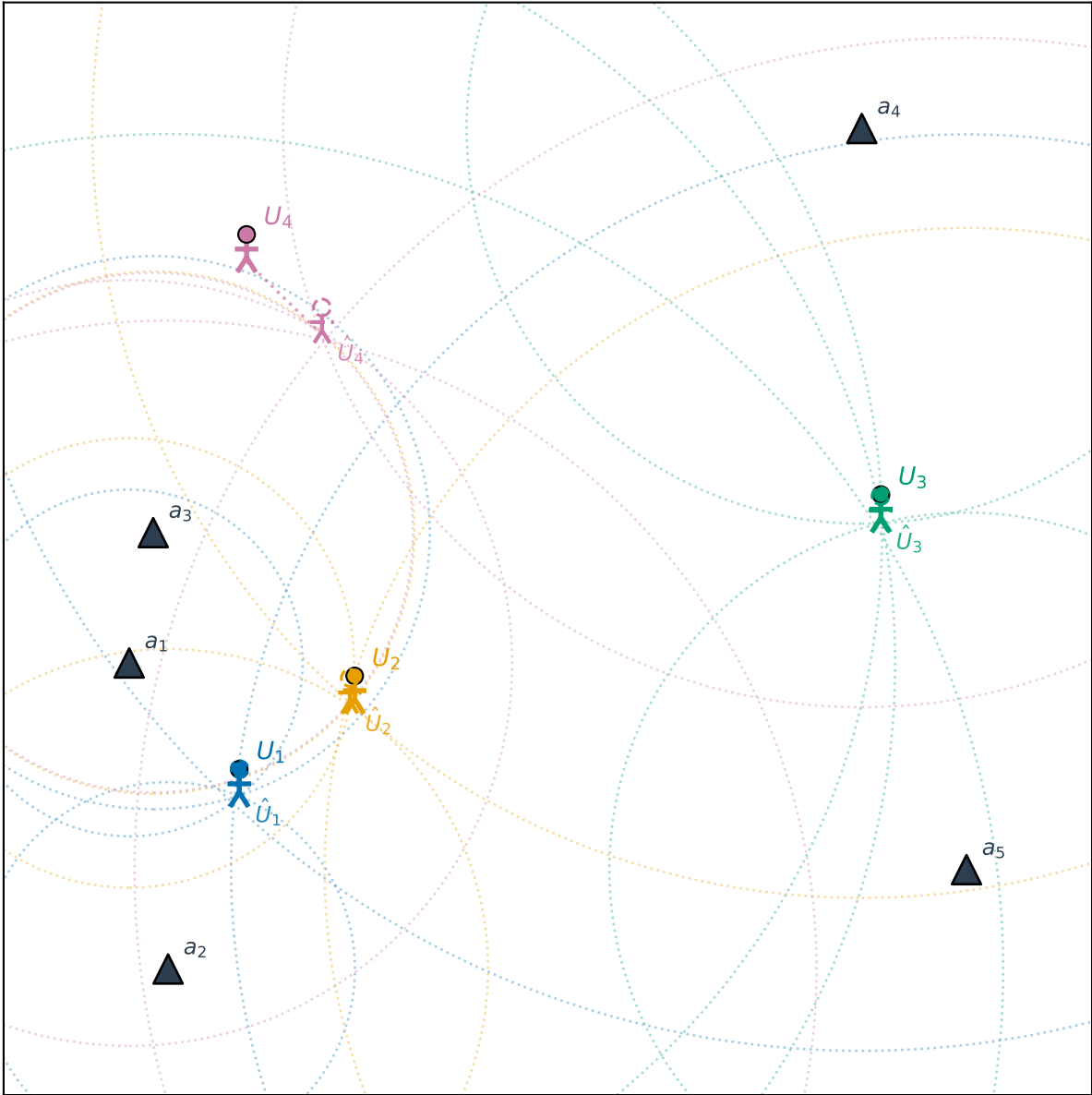
▲ Fixed Antennas (P) □ True User Region □ Recovered User for σ_l

Noise Level 0/5: Noiseless Baseline (0% RMS, SNR = ∞ dB) ($\sigma = 0.0000$)
Tensor Rec. Error: 4.2480e-08

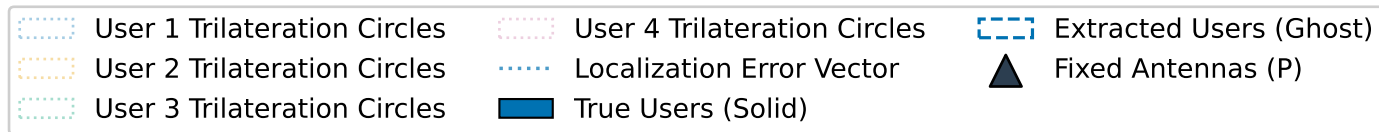
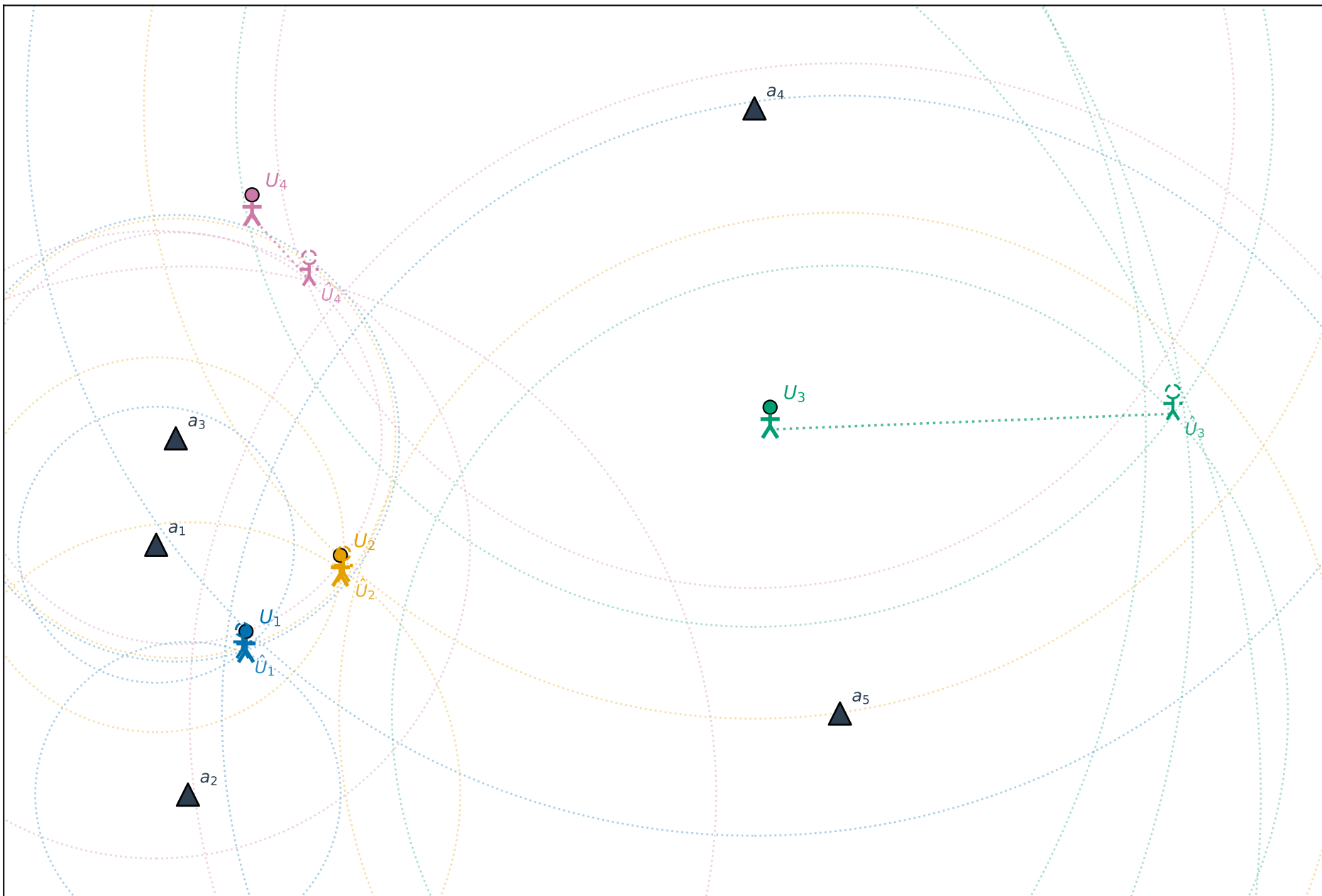


 User 1 Trilateration Circles	 User 4 Trilateration Circles	 Extracted Users (Ghost)
 User 2 Trilateration Circles	 Localization Error Vector	 Fixed Antennas (P)
 User 3 Trilateration Circles	 True Users (Solid)	

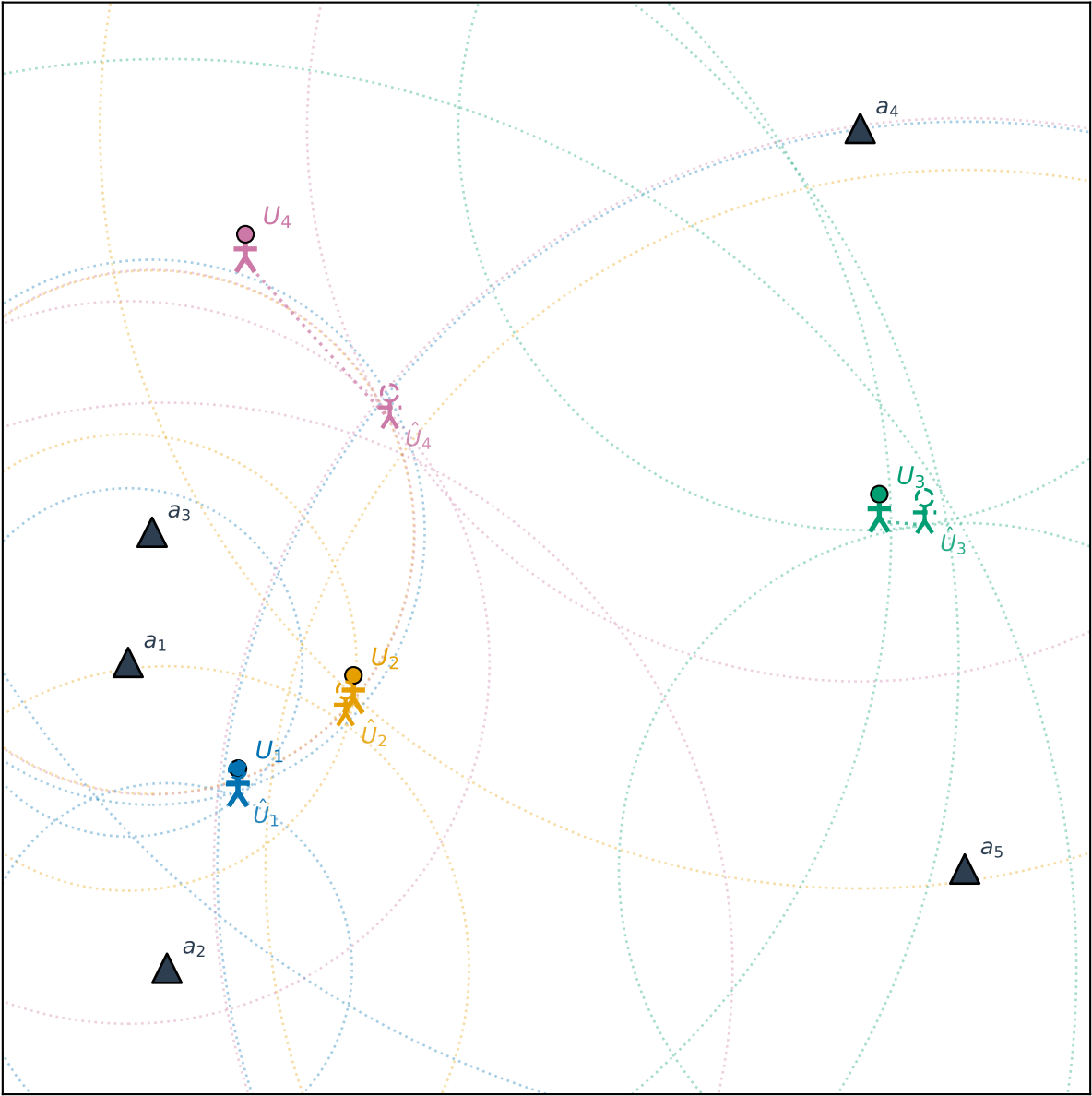
Noise Level 1/5: Relative Noise = 15.0% RMS, SNR = 16.5 dB ($\sigma = 0.0092$)
Tensor Rec. Error: 1.4392e-01



Noise Level 2/5: Relative Noise = 35.0% RMS, SNR = 9.1 dB ($\sigma = 0.0215$)
Tensor Rec. Error: 3.2068e-01

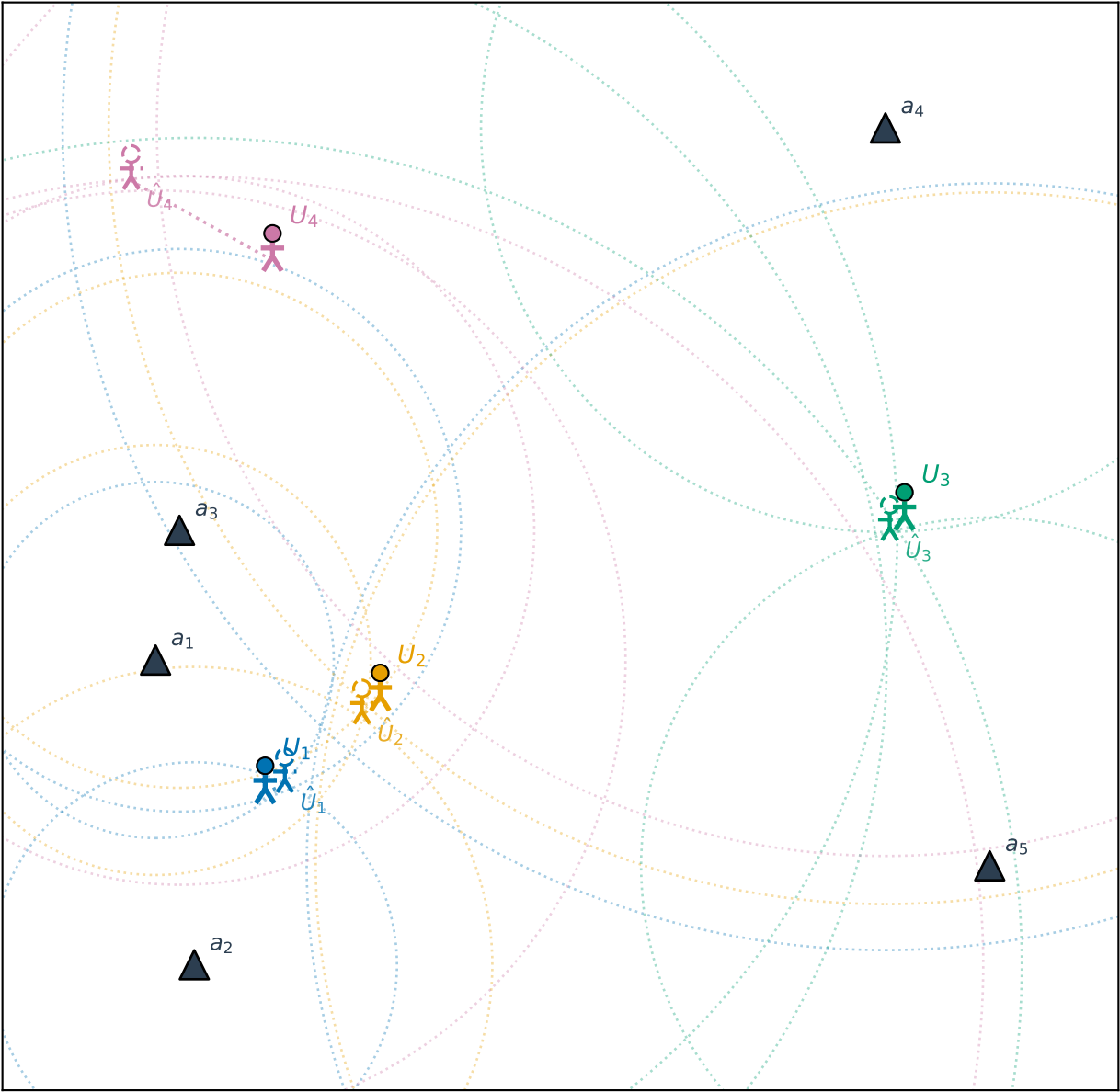


Noise Level 3/5: Relative Noise = 60.0% RMS, SNR = 4.4 dB ($\sigma = 0.0369$)
Tensor Rec. Error: 4.9737e-01



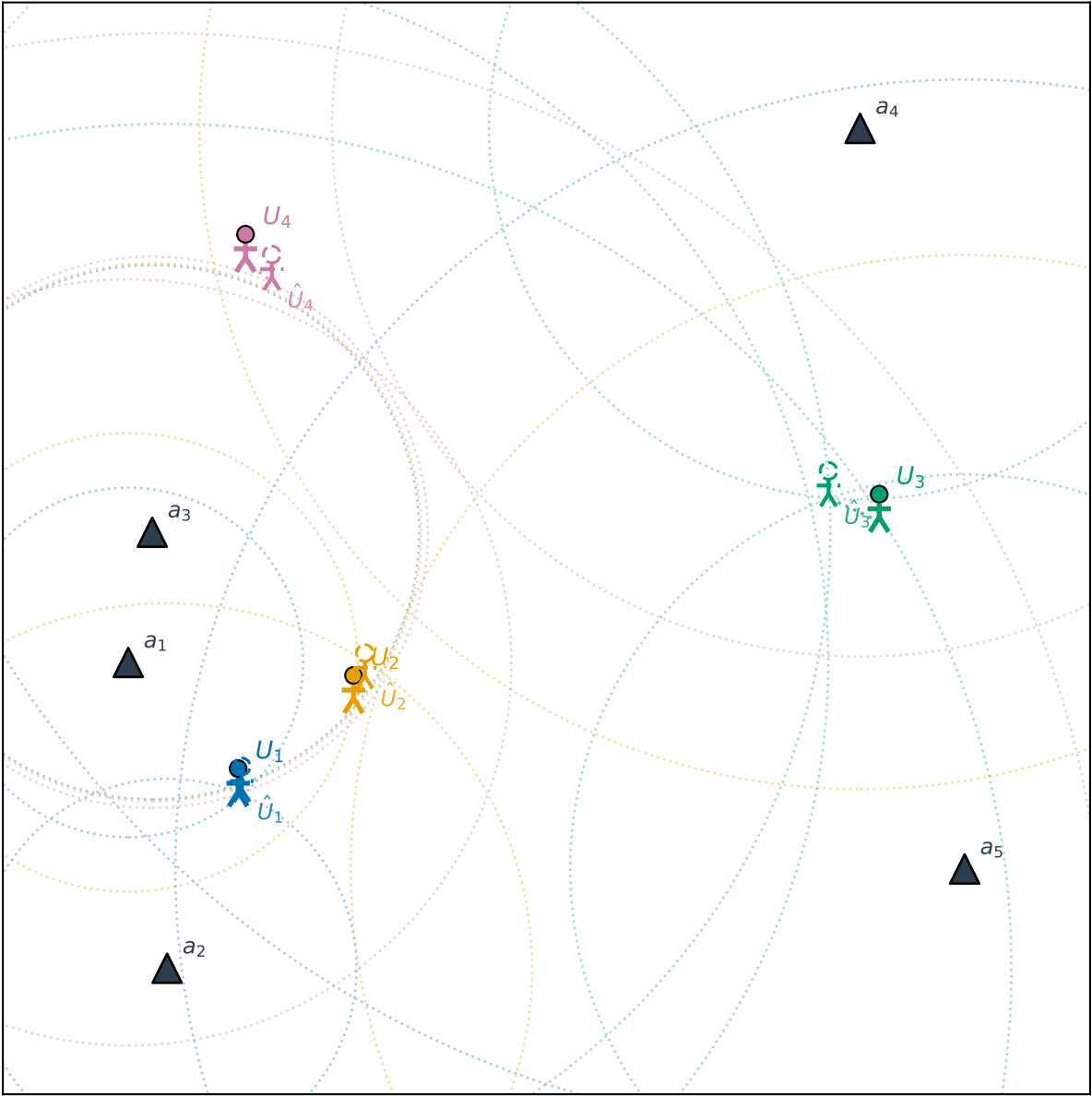
- | | | |
|---|--|---|
|  User 1 Trilateration Circles |  User 4 Trilateration Circles |  Extracted Users (Ghost) |
|  User 2 Trilateration Circles |  Localization Error Vector |  Fixed Antennas (P) |
|  User 3 Trilateration Circles |  True Users (Solid) | |

Noise Level 4/5: Relative Noise = 90.0% RMS, SNR = 0.9 dB ($\sigma = 0.0553$)
Tensor Rec. Error: 6.5143e-01



- | | | |
|---|--|---|
|  User 1 Trilateration Circles |  User 4 Trilateration Circles |  Extracted Users (Ghost) |
|  User 2 Trilateration Circles |  Localization Error Vector |  Fixed Antennas (P) |
|  User 3 Trilateration Circles |  True Users (Solid) | |

Noise Level 5/5: Relative Noise = 120.0% RMS, SNR = -1.6 dB ($\sigma = 0.0737$)
Tensor Rec. Error: 7.4937e-01



 User 1 Trilateration Circles

 User 2 Trilateration Circles

 User 3 Trilateration Circles

 User 4 Trilateration Circles

 Localization Error Vector

 True Users (Solid)

 Extracted Users (Ghost)

 Fixed Antennas (P)